9B Lesson 2

Plant adaptations: roots, stems and leaves

Year 9 Science · 9B Plant growth · Student Book pages 24 to 25

Name _____ Class _____ Date _____

Today I can:

☐ I can describe what plants need from the soil and the air, and how roots, root hair cells and xylem are adapted to supply water and mineral salts.

☐ I can explain how a leaf and its palisade cells are adapted for photosynthesis.

Key words

mineral salts substances dissolved in soil water that plants need to stay healthy, e.g. nitrates

nitrate a mineral salt that supplies nitrogen for making proteins

magnesium an element plants need to make chlorophyll

potassium an element root cells need to absorb water properly

root hair cell a root cell with a long extension that gives a large surface area for absorbing water

xylem hollow tubes made from dead cells that carry water and mineral salts up the plant

wilt to droop because the cells have too little water

palisade cell a leaf cell near the upper surface that is packed with chloroplasts

epidermis the outer layer of cells of a leaf

cuticle the waxy layer on a leaf that stops it losing too much water

surface area the total area of the outside of something

● Core Everyone does this section.

1. Complete the table to show how plants get what they need for growth. Use the word bank: air, soil water, respiration, photosynthesis, making proteins, making chlorophyll, absorbing water, keeping cells firm.

Substance	Where it comes from	How the plant uses it
oxygen		
carbon dioxide		
nitrate salts		
magnesium salts		
potassium salts		
water		

2. Match each part of a leaf to how it helps the plant. Draw a line.

palisade cells	carries water and mineral salts from the roots
phloem	stops the leaf losing too much water
xylem	allow gases to enter and leave the leaf
stomata	contain many chloroplasts, near the top where there is most light
cuticle	carries dissolved sugars from the leaf to other parts of the plant

3. Complete the sentences about the route of water. Use the word bank.

Water enters the plant through _____ cells, which have a large _____ so they absorb water quickly.

The water travels up the root and stem in _____ vessels, which are hollow tubes made from _____ cells.

If a plant has too little water, its cells sag and the plant _____.

●● Stretch When Core is finished.

S1. Describe how each of these cells is adapted for its function. Say what the cell is like and what that lets it do.

a) xylem cells

b) root hair cells

S2. A plant short of magnesium has yellow leaves and grows slowly. Explain why it grows slowly. Use the words chlorophyll, light and glucose.

S3. Explain why water is important to a plant. Give three different reasons.

●●● Challenge Explain in full sentences.

C1. The Venus flytrap grows in bogs with very few mineral salts in the soil. It catches insects when the trigger hairs in its trap are touched more than once.

a) Suggest why a trigger hair has to be touched more than once before the trap shuts.

b) State what the Venus flytrap gets from the insects, and explain why it might die without them even though it can photosynthesise.

C2. Describe how a leaf is adapted to allow its palisade cells to get all the resources they need for photosynthesis: light, carbon dioxide and water.

My thinking (any language)

Some farmers grow lettuce in water with mineral salts dissolved in it, and no soil at all. Could this work? What would the plants still need, and where would it come from?

Word bank mineral salts · nitrate · magnesium · potassium · root hair cell · xylem · wilt · palisade cell · cuticle · surface area · dead

✂ ---------- cut here

Exit ticket 9B Lesson 2: Plant adaptations: roots, stems and leaves Name _____

1. Name the mineral salt a plant needs to make proteins.

_____

2. How are root hair cells adapted to absorb water?

_____

3. Why do palisade cells contain so many chloroplasts?

_____

Today I can: Goal 1 ☐ Goal 2 ☐